

| Instance        | Design unit |
|-----------------|-------------|
| present_tb      | present_tb  |
| run_test        | present_tb  |
| u_4bit          | present_4b  |
| u_kd            | present_kd  |
| #ALWAYS#22      | present_tb  |
| #INITIAL#57     | present_tb  |
| #vsim_capacity# |             |

| Name        | Now     |
|-------------|---------|
| clk         | 0       |
| rst         | 0       |
| start       | 0       |
| plaintext   | 0000... |
| key         | 0000... |
| done_4bit   | St1     |
| done_kd     | St1     |
| cipher_4bit | 0010... |
| cipher_kd   | 0111... |

| Name        | Type (filter) |
|-------------|---------------|
| #INITIAL#57 | Initial       |
| #ALWAYS#22  | Always        |

```
# [KD] Round 27 : abdacec374380d74 (S-box sel=15)
# [4bit] Round 27 : 65d50da21fbcc09f
# [KD] Round 28 : 83990718ca08ce28 (S-box sel=3)
# [4bit] Round 28 : 6a50663c540d862f
# [KD] Round 29 : f8a24565b1d9e785 (S-box sel=1)
# [4bit] Round 29 : c79b8ff00a48df35
# [KD] Round 30 : 692fa0244c899162 (S-box sel=15)
# [4bit] Round 30 : 4a38c5e00283fba1
# [KD] Round 31 : 25af42e5a0555fba (S-box sel=11)
# [4bit] Round 31 : 38d2f04c34635345
# =====
# Plaintext : 0000000000000000
# Key : 0000000000000000
# -----
# [4-bit Iterative Architecture]
# Ciphertext : 5579c1387b228445
# Expected : 5579c1387b228445
# Result : PASS
# -----
# [Modified Key Dependent Architecture]
# Ciphertext : 48047391ef1488ba
# Note: Different S-box used each round
# Same input gives different output
# Security increased by factor of 2^4
# =====
# >>> TEST 2: PRESENT Spec Vector 4
# --- Round-by-Round Data ---
# [KD] Round 1 : 00000000ffffff (S-box sel=0)
# [4bit] Round 1 : ffffffff00000000
# [KD] Round 2 : 000000ff3f500800 (S-box sel=13)
```

```
Ln#
80      64'h0000000000000000,
81      80'h000000000000000000,
82      64'h5579c1387b228445
83      );
84
85      $display(">>> TEST 2: PRESENT Spec Vector 4");
86      $display("--- Round-by-Round Data ---");
87      run_test(
88          64'hffffffffffffff,
89          80'hffffffffffffff,
90          64'h3333dcd3213210d2
91      );
92
93      $display(">>> TEST 3: Paper Test Vector 3");
94      $display("--- Round-by-Round Data ---");
95      run_test(
96          64'h00000000000003b1e,
97          80'h000000000000050ba282,
98          64'h21eb3cf6625a7db3
99      );
100
101      $display("==== Simulation Complete =====");
102      $display("Simulation Complete");
103      $display("====");
104      $finish;
105      end
106
107      endmodule
```

ModelSim - INTEL FPGA STARTER EDITION 10.5b

File Edit View Compile Simulate Add Transcript Tools Layout Bookmarks Window Help

Layout Simulate ColumnLayout AllColumns

sim - Default

Instance Design unit

- present\_tb present\_tb
- run\_test present\_tb
- u\_4bit present\_4b
- u\_kd present\_kd
- #ALWAYS#22 present\_tb
- #INITIAL#57 present\_tb
- #vsim\_capacity#

Objects

| Name        | Now     |    |
|-------------|---------|----|
| clk         | 0       | Re |
| rst         | 0       | Re |
| start       | 0       | Re |
| plaintext   | 0000... | Pa |
| key         | 0000... | Pa |
| done_4bit   | St1     | Ne |
| done_kd     | St1     | Ne |
| cipher_4bit | 0010... | Ne |
| cipher_kd   | 0111... | Ne |

Processes (Active)

| Name        | Type (filter) |
|-------------|---------------|
| #INITIAL#57 | Initial       |
| #ALWAYS#22  | Always        |

Transcript

```
# [KD] Round 29 : c2587a5502879eb1 (S-box sel=1)
# [4bit] Round 29 : 4784ee2eb015e25b
# [KD] Round 30 : 1991f8e90e77676b (S-box sel=15)
# [4bit] Round 30 : c0d7dc77ad955657
# [KD] Round 31 : 996ccf88c5557f06 (S-box sel=2)
# [4bit] Round 31 : cf0bab962dlbaab3

# Plaintext : 00000000000003ble
# Key : 000000000000050ba282

# [4-bit Iterative Architecture]
# Ciphertext : 21eb3cf6625a7db3
# Expected : 21eb3cf6625a7db3
# Result : PASS

# [Modified Key Dependent Architecture]
# Ciphertext : 778c58e88a14a806
# Note: Different S-box used each round
# Same input gives different output
# Security increased by factor of 2^4

# Simulation Complete

# ** Note: $finish : present_tb.v(104)
# Time: 1085 ps Iteration: 0 Instance: /present_tb
# 1
# Break in Module present_tb at present_tb.v line 104

VSIM 20>
```

1telFPGA\_lite/18.0/present/present\_tb.v (/present\_tb) - Default

| Ln# |                                        |
|-----|----------------------------------------|
| 80  | 64'h0000000000000000,                  |
| 81  | 80'h000000000000000000,                |
| 82  | 64'h5579c1387b228445                   |
| 83  | );                                     |
| 84  |                                        |
| 85  | \$display(">>> TEST 2: PRESENT Spec Ve |
| 86  | \$display("--- Round-by-Round Data --- |
| 87  | run_test(                              |
| 88  | 64'hffffffffffffffff,                  |
| 89  | 80'hffffffffffffffffffff,              |
| 90  | 64'h3333dcd3213210d2                   |
| 91  | );                                     |
| 92  |                                        |
| 93  | \$display(">>> TEST 3: Paper Test Vect |
| 94  | \$display("--- Round-by-Round Data --- |
| 95  | run_test(                              |
| 96  | 64'h00000000000003ble,                 |
| 97  | 80'h000000000000050ba282,              |
| 98  | 64'h21eb3cf6625a7db3                   |
| 99  | );                                     |
| 100 |                                        |
| 101 | \$display("====="                      |
| 102 | \$display(" Simulation Complete");     |
| 103 | \$display("====="                      |
| 104 | \$finish;                              |
| 105 | end                                    |
| 106 |                                        |
| 107 | endmodule                              |

ModelSim - INTEL FPGA STARTER EDITION 10.5b

File Edit View Compile Simulate Add Transcript Tools Layout Bookmarks Window Help

Layout Simulate

ColumnLayout AllColumns

sim - Default

Instance Design unit

- present\_tb present\_tb
- run\_test present\_tb
- u\_4bit present\_4b
- u\_kd present\_kd
- #ALWAYS#22 present\_tb
- #INITIAL#57 present\_tb
- #vsim\_capacity#

Objects

| Name        | Now     | Type |
|-------------|---------|------|
| clk         | 0       | Re   |
| rst         | 0       | Re   |
| start       | 0       | Re   |
| plaintext   | 0000... | Pa   |
| key         | 0000... | Pa   |
| done_4bit   | St1     | Ne   |
| done_kd     | St1     | Ne   |
| cipher_4bit | 0010... | Ne   |
| cipher_kd   | 0111... | Ne   |

Processes (Active)

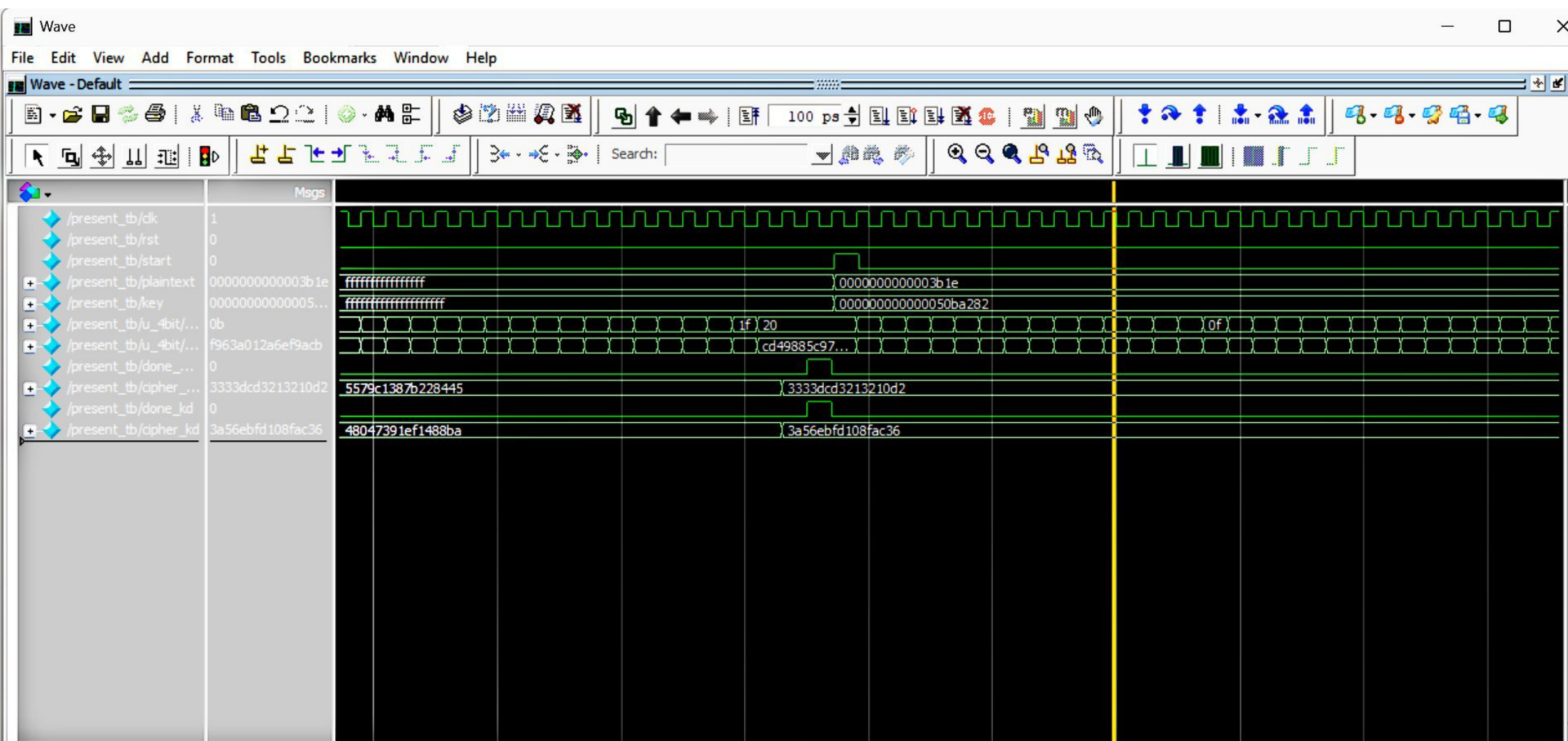
| Name        | Type (filter) |
|-------------|---------------|
| #INITIAL#57 | Initial       |
| #ALWAYS#22  | Always        |

Transcript

```
# [KD] Round 30 : 7604210c020b0a2e (S-box sel=13)
# [4bit] Round 30 : 0f92b6866bcfb5d9
# [KD] Round 31 : c42cbf72a6811d51 (S-box sel=4)
# [4bit] Round 31 : cd49885c973calb5
# =====
# Plaintext : ffffffffffffffff
# Key : ffffffffffffffff
# =====
# [4-bit Iterative Architecture]
# Ciphertext : 3333dcd3213210d2
# Expected : 3333dcd3213210d2
# Result : PASS
# =====
# [Modified Key Dependent Architecture]
# Ciphertext : 3a56ebfd108fac36
# Note: Different S-box used each round
# Same input gives different output
# Security increased by factor of 2^4
# =====
# >>> TEST 3: Paper Test Vector
# --- Round-by-Round Data ---
# [KD] Round 1 : 00060001ffff1ffc (S-box sel=0)
# [4bit] Round 1 : fff8fff20008000e
# [KD] Round 2 : 81fe17114eee3600 (S-box sel=12)
# [4bit] Round 2 : 68eealeeb7103011
# [KD] Round 3 : 6ab11a4409822a7e (S-box sel=0)
# [4bit] Round 3 : 9f9d1837fa48d12a
# [KD] Round 4 : 22c6a59cbdeee436 (S-box sel=6)
# [4bit] Round 4 : 229d3cfef65alae0
```

itelfPGA\_lite/18.0/present/present\_tb.v (/present\_tb) - Default

| Ln# | Code                                    |
|-----|-----------------------------------------|
| 80  | 64'h0000000000000000,                   |
| 81  | 80'h000000000000000000,                 |
| 82  | 64'h5579c1387b228445                    |
| 83  | );                                      |
| 84  |                                         |
| 85  | \$display(">>> TEST 2: PRESENT Spec Ver |
| 86  | \$display("--- Round-by-Round Data ---  |
| 87  | run_test(                               |
| 88  | 64'hffffffffffffffff,                   |
| 89  | 80'hffffffffffffffff,                   |
| 90  | 64'h3333dcd3213210d2                    |
| 91  | );                                      |
| 92  |                                         |
| 93  | \$display(">>> TEST 3: Paper Test Vecto |
| 94  | \$display("--- Round-by-Round Data ---  |
| 95  | run_test(                               |
| 96  | 64'h00000000000003ble,                  |
| 97  | 80'h000000000000050ba282,               |
| 98  | 64'h21eb3cf6625a7db3                    |
| 99  | );                                      |
| 100 |                                         |
| 101 | \$display("=====                        |
| 102 | \$display(" Simulation Complete");      |
| 103 | \$display("=====                        |
| 104 | \$finish;                               |
| 105 | end                                     |
| 106 |                                         |
| 107 | endmodule                               |



present

project Navigator Files

- sbox.v
- pbox.v
- key\_transform.v
- present\_4bit.v
- present\_kd.v
- present\_tb.v
- kd\_sbox.v

Tasks Compilation

| Task                   |
|------------------------|
| Compile Design         |
| Analysis & Synthesis   |
| Edit Settings          |
| View Report            |
| Analysis & Elaboration |
| Partition Merge        |

Table of Contents

- Flow Summary
- Flow Settings
- Flow Non-Default Global Settings
- Flow Elapsed Time
- Flow OS Summary
- Flow Log
- Analysis & Synthesis
- Fitter
- Assembler
- Timing Analyzer
- EDA Netlist Writer
- Flow Messages
- Flow Suppressed Messages

Flow Summary

<<Filter>>

|                                    |                                             |
|------------------------------------|---------------------------------------------|
| Flow Status                        | Successful - Sun Jun 28 20:22:59 2026       |
| Quartus Prime Version              | 18.0.0 Build 614 04/24/2018 SJ Lite Edition |
| Revision Name                      | present                                     |
| Top-level Entity Name              | present_4bit                                |
| Family                             | MAX 10                                      |
| Device                             | 10M50DAF484C7G                              |
| Timing Models                      | Final                                       |
| Total logic elements               | 320 / 49,760 ( < 1 % )                      |
| Total registers                    | 218                                         |
| Total pins                         | 212 / 360 ( 59 % )                          |
| Total virtual pins                 | 0                                           |
| Total memory bits                  | 0 / 1,677,312 ( 0 % )                       |
| Embedded Multiplier 9-bit elements | 0 / 288 ( 0 % )                             |
| Total PLLs                         | 0 / 4 ( 0 % )                               |
| UFM blocks                         | 0 / 1 ( 0 % )                               |
| ADC blocks                         | 0 / 2 ( 0 % )                               |

IP Catalog

- Installed IP
  - Project Directory
    - No Selection Available
  - Library
    - Basic Functions
    - DSP
    - Interface Protocols
    - Memory Interfaces and Controllers
    - Processors and Peripherals
    - University Program
  - Search for Partner IP

+ Add...

All Find... Find Next

Messages

| Type | ID     | Message                                                               |
|------|--------|-----------------------------------------------------------------------|
| Info | >      | Quartus Prime EDA Netlist Writer was successful. 0 errors, 2 warnings |
| Info | 293000 | Quartus Prime Full Compilation was successful. 0 errors, 15 warnings  |

System (8) Processing (132)

Starts a new compilation

100% 00:01:37

## Netlist Navigator

Fit in Window - Ctrl+Alt+W

- present\_4bit
  - State Machines
  - Instances
    - key\_transform:u1
    - pbox:u\_pbox
      - Ports
        - Input
          - in[63..0]
        - Output
          - out[3..0]
      - sbox:u0
      - sbox:u1
      - sbox:u2
      - sbox:u3
        - Primitives
        - Ports
          - Input
          - Output
      - sbox:u4
      - sbox:u5
      - sbox:u6
      - sbox:u7
      - sbox:u8
      - sbox:u9
      - sbox:u10
      - sbox:u11
      - sbox:u12
      - sbox:u13
      - sbox:u14

